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MID-SEMESTER EXAMINATION, June 2026
SUMMER SEMESTER (CBCS)

Degree: ...B.Tech...CSIOT / CSDA

Semester: 3rd Sem

Course Title: Digital Logic Design

Course Code: CIEPC03

Duration: 1:30 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1 a	Represent (-29.625) in 2's complement form.	1.5	CO1
1 b	Find BCD sum of 289 and 714	1.5	CO1
2 a	Implement using only NOR gates: $A'.B + A.B'.C$	1.5	CO1
2 b	Design a BCD adder using parallel adders.	1.5	CO1
3 a	Design an EXOR gate using only NOR gates.	1.5	CO2
3 b	Design a 3 bit Gray to Binary code converter.	1.5	CO1
4 a	Design a 8 to 1 multiplexer using universal gates only.	1.5	CO2
4 b	Design a full subtractor using logic gates.	1.5	CO2
5 a	Implement a 16:1 MUX using five 4:1 MUX.	1.5	CO2
5 b	Minimize the expression using K-map and implement with NAND gates: $F(A, B, C, D) = \sum m(0, 2, 7, 8, 10) + d(5, 13, 15)$	1.5	CO2